Optimal Dynamic Pricing With Preferences for Novelty

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Motivation

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Research Question

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Supply Setting

- Monopolistic firm producing a single product j
- Firm is profit maximizing, setting prices according to that objective
- A consumer is born into the world with a type $r \in \{H, L\}$, determining how much utility they derive from switching and an initial state $h_0 \in \{0, j\}$
- Consumers live for two periods, purchasing the inside good or opting for the outside good in each period
- The firm knows initial states and types, setting prices to individual consumers by type and history

Consumer's Problem

$$V_{jt}^{inside} = \underbrace{\beta}_{\text{Constant}} + \overbrace{\gamma_r \mathbb{1}\{h=0\}}^{\text{Utilty from Switching}} - \underbrace{\alpha p_t(h)}_{\text{Price Disutility}}, \quad V_{0t}^{Outside} = \underbrace{\gamma_r \mathbb{1}\{h=j\}}_{\text{Utility from Switching}}$$

$$v_t(x) = V_{jt}^{inside} - V_{0t}^{outside} = \beta + \underbrace{\gamma[\mathbb{1}\{h=0\} - \mathbb{1}\{h=j\}]}_{q(h)} - \alpha p_t(h)$$

- Prices, set by the firm, and switching utility depend on history $h = x_{t-1}$
- Utility for the outside good is normalized to zero

Firm's Problem

$$\pi_t(h) = \max_{p_t(h)} s_t(h)[p_t(h) - c]$$

$$\Pi(h) = \underbrace{s_1(h)(p_1(h) - c)}_{\text{Margin in Period 1}} + \underbrace{\delta\{s_1(h)\pi_2(j) + (1 - s_1(h))\pi_2(0)\}}_{\text{Continuation Value}}$$

- Firm profits are comprised of period one margins, given history of consumption, and the value of that consumer in the different states in period two
- The firm will set prices according to consumer type and history to maximize expected profit

Optimal Terminal Price and Profit

$$\frac{\partial \pi_2(h_1)}{\partial p_2(h_1)} = s_2(h_1) + (p_2 - c)\frac{\partial s_2(h_1)}{\partial p_2(h_1)}$$

$$p_2^*(h_1) = c + \frac{1}{\alpha(1 - s_2^*(h_1))}$$

$$\pi_2^*(p_2^*(h_1); h_1) = \frac{s_2^*(h_1)}{\alpha(1 - s_2^*(h_1))}$$

- Optimal period two price is cost minus a demand markup term
- Period two profit at optimal prices is the odds of purchase, scaled by the inverse price sensitivity

First Period Optimization

$$\frac{d\Pi}{dp_1(h_0)} = s_1 + \frac{\partial}{\partial p_1(h)} s_1(h_0)[p_1(h_0) - c + \delta(\pi_2(j) - \pi_2(0))]$$

$$p_1^*(h_0) = c + \frac{1}{\alpha(1 - s_1^*(h_0))} - \delta(\pi_2(j) - \pi_2(0))$$

$$\Pi^*(h) = \delta\pi_2(0) + \frac{s_1^*(h_0)}{\alpha(1 - s_1^*(h_0))}$$

Optimal Profit Given Switching Type

$$\frac{d\Pi}{d\gamma} = \frac{q(h)s_1^*(h) - \delta s_1^*(h)s_2^*(j) + \delta(1 - s_1^*(h))s_2^*(0)}{\alpha}$$

- There are two interpretations:
  1. If $h_0 = j$, $q(h) < 0$, leading us to ambiguous signage as we raise profits from initially outside consumers but see opposing forces for the initially inside: leave in period one, return in period two
  2. If $h_0 = 0$, $q(h) > 0 \Rightarrow \frac{d\Pi}{d\gamma} < 0$, meaning that profits decrease as γ increases
- Points to firms potentially reaping great benefit from knowing consumer types and histories

Pricing Simulations Specification

Parameter	Description	Value
β	Inside-good intercept	2
α	Price sensitivity	1
δ	Discount factor	0.92
c	Marginal cost	0
γ	Switching preference	$[-3, 3]$
Initial Consumption State		$h_0 \in \{0, j\}$
Feasible Prices		$p_t(h) \in \mathbb{R}$

- Allow for negative pricing
- Firm has knowledge of both type and history, allowing for targeted pricing
- Four consumption histories possible $(h_0, h_1) \in \{(0,0), (0,j), (j,j), (j,0)\}$

Shares and Elasticities

$$s_t(h) = \frac{e^{v_t(h)}}{1 + e^{v_t(h)}}, \; s_{0t}(h) = 1 - s_t(h)$$

$$\frac{\partial s_t(h)}{\partial p_t(h)} = -\alpha s_t(h)(1 - s_t(h))$$

$$\frac{\partial s_t(h)}{\partial \gamma} = q(h) s_t(h)(1 - s_t(h))$$